Temperature Sensor Project (Arduino-Based)

Project Overview

In this project, students will design and build a temperature monitoring system using an Arduino Uno, an analog and a digital temperature sensor, and an LCD display. The system will be assembled on a breadboard and will demonstrate how sensors collect and transmit data. Students will program the Arduino to read, process, and display temperature values using ADC and digital communication.

Project Objectives

The objective of this project is to help students understand the differences between analog and digital sensors while developing skills in circuit assembly, programming, and data interpretation. Students will use ADC to read analog signals, apply I2C communication for the LCD, and convert temperature readings into multiple units, while building teamwork and problem-solving skills.

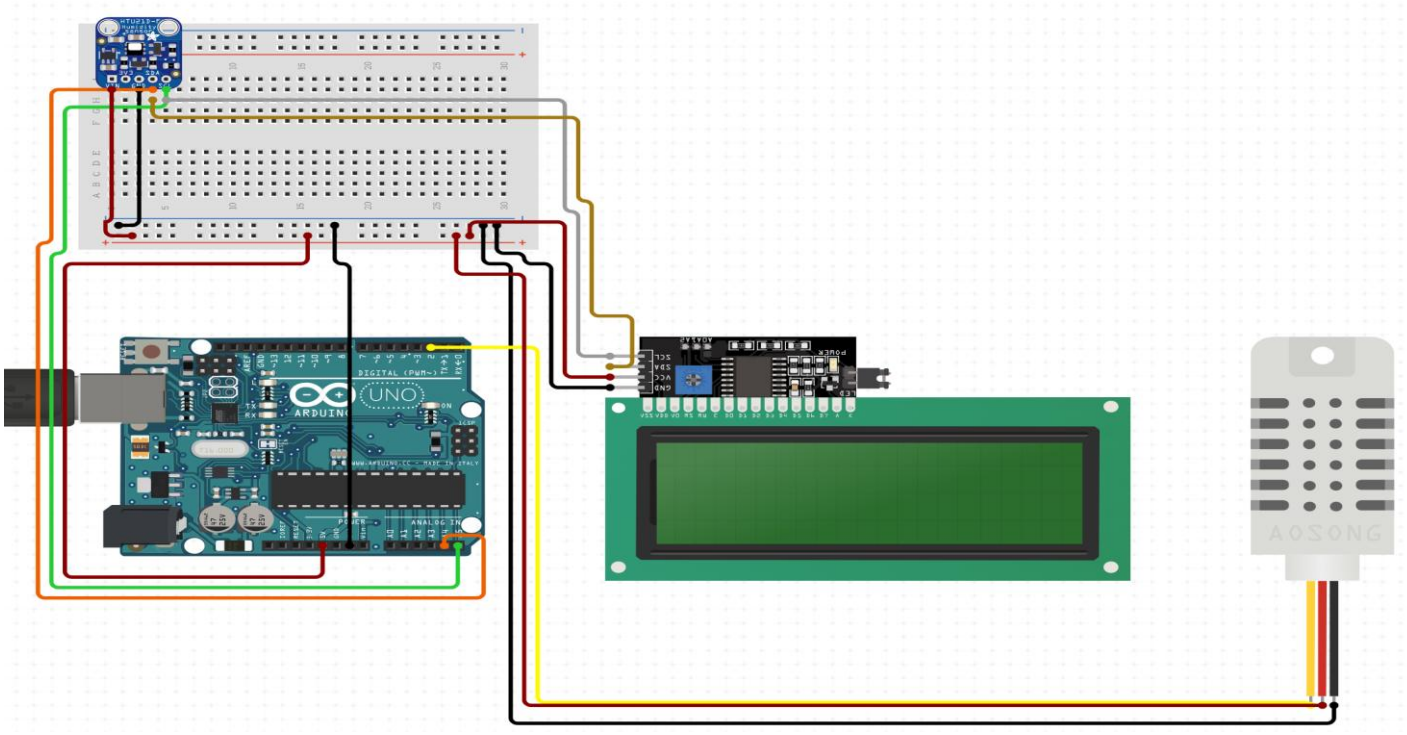
Group Structure

Students will work in small groups of 3 to 4, sharing responsibilities across design, assembly, coding, testing, and documentation. Effective communication and collaboration are essential to ensure the system is successfully integrated.

System Description

The system collects temperature data from both an analog and a digital sensor. The analog sensor produces a voltage that varies with temperature, which is read by the Arduino using its ADC and converted into a Celsius value. The digital sensor provides calibrated temperature readings directly through a sensor library.

The Arduino then processes both readings and converts them into Fahrenheit and Kelvin using standard formulas. The final values are displayed on an LCD screen, which communicates with the Arduino via the I2C protocol, reducing wiring and simplifying the overall circuit design.



Components and Materials
Each group will require the following components:

- **Arduino Uno** – microcontroller board for processing data
- **Breadboard** – for assembling the circuit without soldering
- **16x2 LCD Display with I2C module** – for displaying temperature readings
- **Analog temperature sensor (e.g., LM35)** – outputs temperature as a voltage signal
- **Digital temperature sensor (e.g., DHT11 or DHT22)** – provides calibrated digital temperature data
- **Jumper wires (male-to-male)** – for making connections between components
- **USB cable** – for powering and programming the Arduino
- **Resistors (if required)** – for proper sensor operation (e.g., pull-up resistor for DHT sensor)

Programming Requirements
Students are required to develop an Arduino sketch that reads input from both sensors, processes the data, and displays the results. The program should continuously update the temperature readings and clearly present the values from both sensors on the LCD screen. Proper use of functions, libraries, and clear code structure is expected.

Expected Output
The LCD display should show temperature readings from both the analog and digital sensors, including values in Celsius, Fahrenheit, and Kelvin. The output should be formatted clearly so that users can easily distinguish between the two sensor readings.

Step-by-Step Build Guide

1. **Power Setup**
   Connect the Arduino Uno to the breadboard by linking the 5V pin to the positive rail and the GND pin to the negative rail. This will supply power to all components.
2. **Analog Temperature Sensor (LM35)**
   Place the sensor on the breadboard. Connect VCC to 5V, GND to ground, and the output pin to analog pin A0 on the Arduino. Upload a simple test sketch to verify readings.
3. **Digital Temperature Sensor (DHT11/DHT22)**
   Connect the sensor's VCC and GND to the power rails. Connect the data pin to a digital pin (e.g., pin 2). Add a pull-up resistor if required. Test using a suitable library.
4. **LCD Display (I2C Module)**
   Connect VCC and GND to the power rails. Connect SDA to A4 and SCL to A5 on the Arduino Uno. Run an I2C scanner or LCD test code to confirm it is working.
5. **System Integration**
   Combine all components into one circuit. Update your Arduino code to read both sensors, process the data, and display the results on the LCD.
6. **Testing and Verification**
   Power the system and observe the output. Check that both sensors display correct temperature values and that the LCD updates continuously. Fix any wiring or code issues if needed.